EXPERIMENT-2

AIM:

To convert the ER diagram of an Indian e-commerce platform into a relational schema and implement it using MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints.

SOURCE CODE:

```
CREATE DATABASE EcommerceDB;
USE EcommerceDB;

CREATE TABLE Customer (
CustomerID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Email VARCHAR(100) UNIQUE NOT NULL,
Phone VARCHAR(15),
DOB DATE
);

CREATE TABLE Customer_Address (
    CustomerID INT,
    Address VARCHAR(200),
```

```
    PRIMARY KEY (CustomerID, Address),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE Seller (
    SellerID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Phone VARCHAR(15),
    EMAIL VARCHAR (100) UNIQUE NOT NULL
);

CREATE TABLE Category (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR (50) NOT NULL UNIQUE
);

CREATE TABLE Product (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR (100) NOT NULL,
    Price DECIMAL (10,2) NOT NULL,
    Specifications VARCHAR (200),
```

```
    SellerID  INT NOT NULL,
    CategoryID INT,
    FOREIGN KEY (SellerID)
        REFERENCES Seller (SellerID)
        ON DELETE CASCADE,
    FOREIGN KEY (CategoryID)
        REFERENCES Category (CategoryID)
        ON DELETE SET NULL
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    TotalAmount DECIMAL (10,2) NOT NULL,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID)
        REFERENCES Customer (CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE OrderItem(
    OrderIDINT,
    ProductID INT,
    Quantity INT NOT NULL,
```

```
    UnitPrice DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID)
        REFERENCES Orders (OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY,
    Amount DECIMAL (10,2) NOT NULL,
    PaymentMode VARCHAR (30) NOT NULL,
    PaymentDate DATE,
    Status VARCHAR (20),
    OrderID INT NOT NULL,
    FOREIGN KEY (OrderID)
        REFERENCES Orders (OrderID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Delivery (
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    Status VARCHAR(20),
    TrackingNo VARCHAR(50) UNIQUE,
    OrderID INT NOT NULL,
    FOREIGN KEY (OrderID)
        REFERENCES Orders (OrderID)
        ON DELETE CASCADE
);

CREATE TABLE Electronics (
    ProductID INT PRIMARY KEY,
    Warranty VARCHAR(50),
    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Clothing (
    ProductID INT PRIMARY KEY,
    Size VARCHAR(20),
    Material VARCHAR(50),
    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Books (
    ProductID INT PRIMARY KEY,
    Author VARCHAR (100),
    ISBN VARCHAR (20) UNIQUE,
    FOREIGN KEY (Product ID)
        REFERENCES Product (Product ID)
        ON DELETE CASCADE
);
-- SAMPLE DATA
INSERT INTO Customer
VALUES
(1, 'Rahul', 'rahul@gmail.com', '9876543210', '2005-05-10');

INSERT INTO Customer_Address
VALUES
(1, 'Delhi');

INSERT INTO Seller
VALUES
(101, 'TechStore', '9876500000', 'techstore@gmail.com');

INSERT INTO Category
VALUES
(201, 'Electronics');
```

```
INSERT INTO Product
 VALUES
 (301, 'LAPTOP', 55000.00, '16GB RAM, 512GB SSD', 101, 201);

INSERT INTO Order
 VALUES
 (401, '2026-08-13', 55000.00, 1);

INSERT INTO OrderItem
 VALUES
 (401, 301, 1, 55000.00);

INSERT INTO Payment
 VALUES
 (601, '2026-08-16', 'S4
 (501, 55000.00, 'UPI', '2026-08-13', 'Paid', 401);

INSERT INTO Delivery
 VALUES
 (601, '2026-08-16', 'Shipped', 'TRK12345', 401);

INSERT INTO Electronics
 VALUES
 (301, '2 Years');
```

```
INSERT INTO Product
 VALUES
(302, 'Mobile', 25000.00, '8GB RAM', 999, 201);

INSERT INTO OrderItem
 VALUES
(999, 301, 1, 55000.00);
```

OUTPUT

☒ Error Code: 1452. Cannot add or update a child row:
a foreign key constraint fails (`ecommercedb`.`Product`,
CONSTRAINT `Product_ibfK_1` FOREIGN KEY (`SellerID`)
REFERENCES `seller` (`sellerID`) ON DELETE CASCADE)

☒ Error Code: 1452. Cannot add or update a child row:
a foreign key constraint fails (`ecommercedb`.`Orderitem`,
CONSTRAINT `orderitem_ibfK_1` FOREIGN KEY (`Order ID`)
REFERENCES `Orders` (`Order ID`) ON DELETE CASCADE)